PICOLINO CPC / REV A0
Pico 2 ROM / CH376 USB / W5100S Ethernet / PIC mouse / RP5C01 RTC
Prototype design — bench validation and RP2350 ROM firmware required

CPC bus, address decoding and power

File: bus-power.kicad_sch

Pico 2 ROM interface and level translation

File: pico-rom.kicad_sch

W5100S Ethernet and PIC PS/2 mouse

File: ethernet-mouse.kicad_sch

RP5C01 RTC and CR2032 battery backup

File: rtc.kicad_sch

Sheet: /
File: picolino-cpc.kicad_sch

Title: Picolino CPC — Pico 2 multifunction expansion

Size: A4	Date: 2026-10-07	**Rev: A0**
KiCad E.D.A. 10.0.6		Id: 1/5

PICOLINO CPC / REV A0 — BUS, DECODE & POWER

FE80/FE81 USB | FD20–FD23 Ethernet | FD10 mouse
FD14 RTC data (low nibble), FD15 RTC index

J1 pin 40 BUSRESET differs from pin 41 RESET.
J2 uses the reference parallel CH376 module on a short cable.
Its USB host socket is on the module, not this carrier.

JP1 selects the board supply; never bridge all three pins.
External input: regulated 5V only; common CPC ground. Power CPC before external board supply.
Check power budget before selecting CPC power; external 5V is preferred for USB-drive bring-up.

J1
CPC 50-pin expansion

U2
GAL20V8B / picolino.jed

U3
74HCT21

J2
CH376 parallel / short ribbon

JP1
POWER: 1 CPC / 2 BOARD / 3 EXT

J3
EXT regulated 5V

R1
1k

D2
USB activity

F1
USB 500mA hold

D1
1N5817

C1
100nF

C2
100nF

C3
100nF

C4
100uF / 10V

C5
47uF / 10V

H1
M3 NPTH

H2
M3 NPTH

H3
M3 NPTH

H4
M3 NPTH

PWR_FLAG
#FLG1

PWR_FLAG
#FLG2

PWR_FLAG
#FLG3

PWR_FLAG
#FLG4

PWR_FLAG
#FLG5

PWR_FLAG
#FLG6

Prototype: verify timing and power on hardware

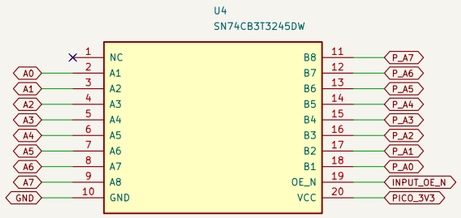
Sheet: /CPC bus, address decoding and power/
File: bus-power.kicad_sch

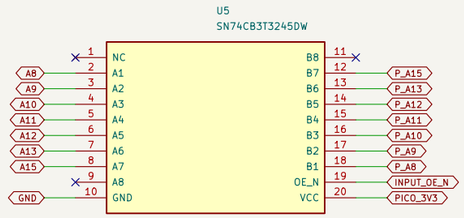
Title: Picolino CPC / bus-power

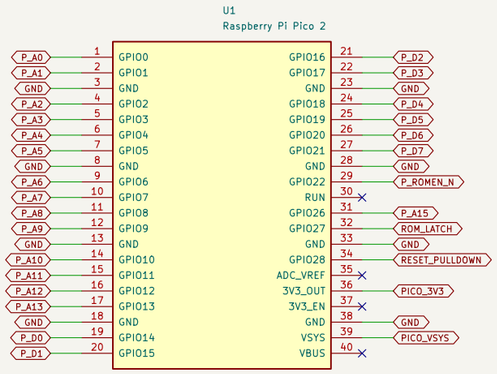
Size: A3	Date: 2026-10-07	**Rev: A0**
KiCad E.D.A. 10.0.6		Id: 2/5

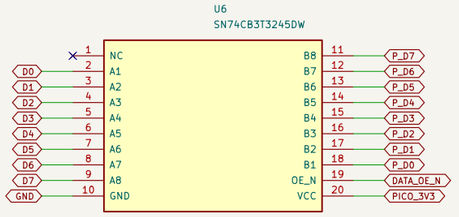
PICO 2 / RP2350 — ROM INTERFACE

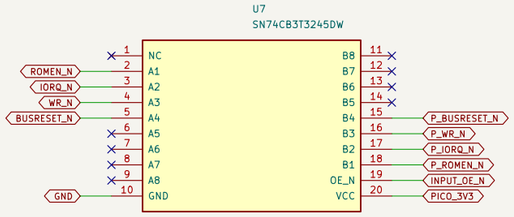
3.3V B-side signals have P_ prefix. All CPC-facing GPIOs use level translation.
Input switches enable when CPC 5V is present; JP2 separately isolates the data bus and controls ROMDIS.



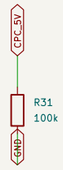








Native RP2350 monitor: keep JP2 open.
ROM-serving firmware and timing remain to be validated.

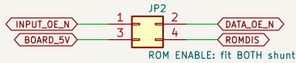


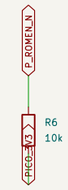










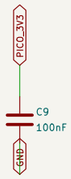


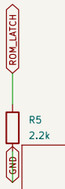
U4–U7 each require local 100nF. GPIO0/1 are address inputs: UART stdio must stay disabled.



















Prototype: verify timing and power on hardware

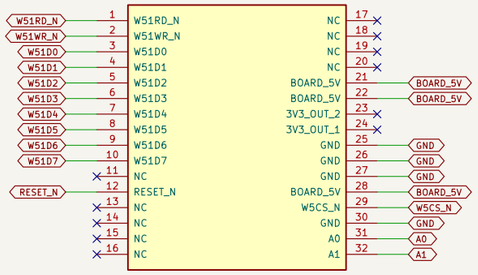
Sheet: /Pico 2 ROM interface and level translation/
File: pico-rom.kicad_sch

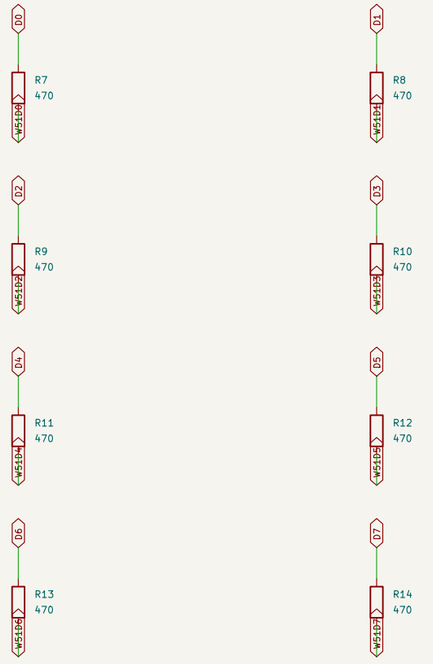
Title: Picolino CPC / pico-rom

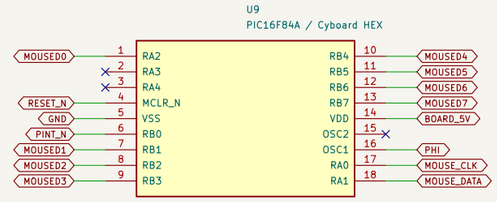
Size: A3	Date: 2026-10-07	**Rev: A0**
KiCad E.D.A. 10.0.6		Id: 3/5

ETHERNET & MOUSE — CYBOARD-COMPATIBLE INTERFACES

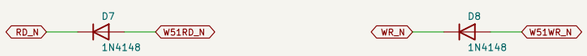
W5100S parallel registers: FD20-FD23. PIC mouse read: FD10.
W5100S module pins 23/24 are its 3.3V outputs and are left unconnected.

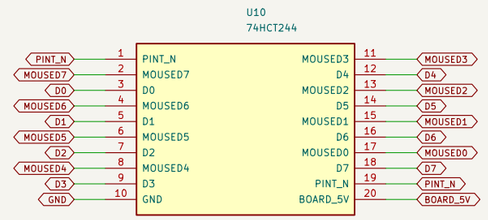


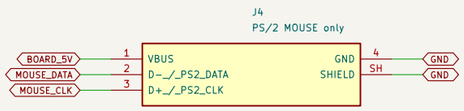


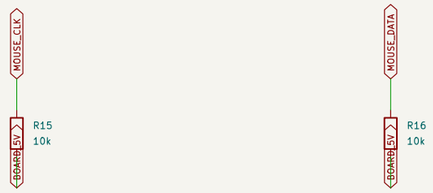


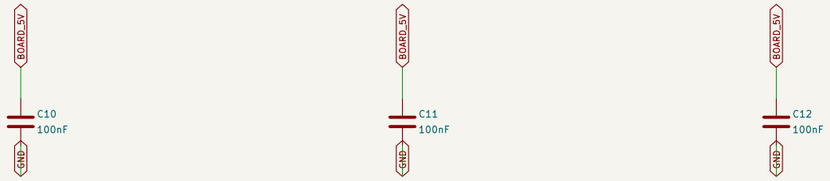
U8 pin numbering and geometry copied from Cyboard module footprint.
Do not fit a SPI-only WIZ810Sio module.











PIC RA0 = mouse clock; RA1 = mouse data.
PIC RA2 carries response D0; retain buffer mapping.
This is not a USB-HID mouse host.

Prototype: verify timing and power on hardware

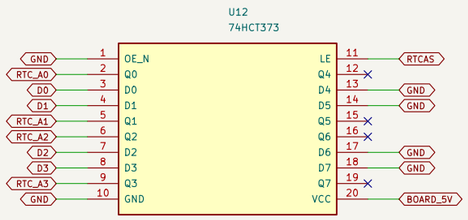
Sheet: /W5100S Ethernet and PIC PS/2 mouse/
File: ethernet-mouse.kicad_sch

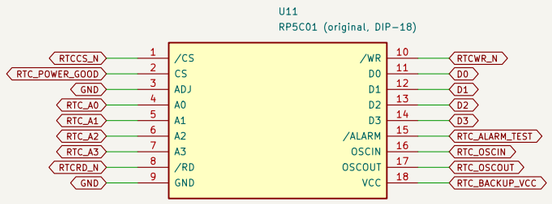
Title: Picolino CPC / ethernet-mouse

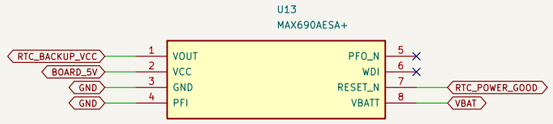
Size: A3	Date: 2026-10-07	**Rev: A0**
KiCad E.D.A. 10.0.6		Id: 4/5

RP5C01 RTC — 4-BIT REGISTERS, CR2032 BACKUP

U11 pin 1 /CS: GAL decode. Pin 2 CS: main-power-good (LOW in backup).
FD15 selects an index; FD14 reads/writes the low nibble. This differs from DS12887 software.

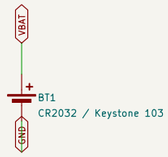




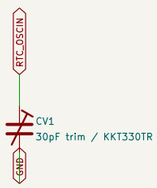


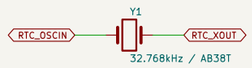
Oscillator follows original Ricoh RP5C01 databook page 3-6, including 56pF on OSCOUT.
Crystal load and trimmer require calibration. Keep oscillator components together, away from digital clocks.
MAX690A powers only U11 from CR2032. WDI is intentionally open; PFI grounded; PFO unused.





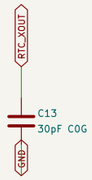


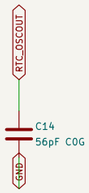




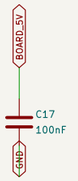






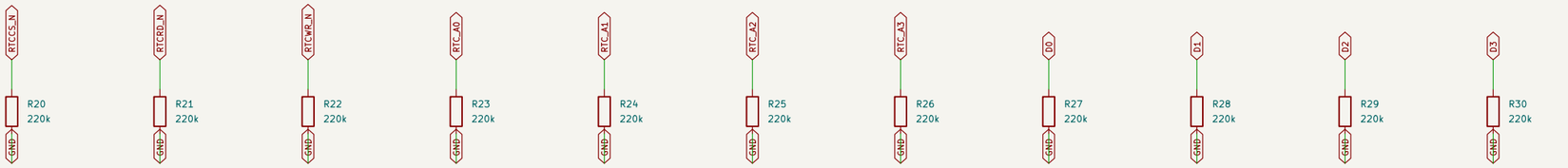








R20-R30 bias RTC inputs during backup. Measure battery current, time retention and supply switching on the prototype.



Prototype: verify timing and power on hardware

Sheet: /RP5C01 RTC and CR2032 battery backup/
File: rtc.kicad_sch

Title: Picolino CPC / rtc

Size: A3	Date: 2026-10-07	**Rev: A0**
KiCad E.D.A. 10.0.6		Id: 5/5